NBA

A RESEARCH PRESENTATION

OPTIMIZATION METHODS II

NBA EXPANSION

Two optimization models for choosing the next two franchises, weighing **travel**, fairness & **market potential** across six candidate cities.

PRESENTED BY

Prince Chinyama

Kamalesh Kumar

SOURCE PAPER AUTHORS

Hassanzadeh, Davari & Goossens

BY THE NUMBERS SEC. 01

06 cities

CANDIDATES

Seattle · Las Vegas · Montreal · Vancouver · Tampa · Mexico City

15 scenarios

CITY-PAIR COMBINATIONS

Every two-city pairing optimized end-to-end

02 frameworks

OPTIMIZATION MODELS

Min-distance · Nash bargaining

32 teams

POST-EXPANSION LEAGUE

First league growth since 2004

INTRODUCTION

30 TEAMS FOR 20 YEARS. TWO MORE ARE COMING.

The league last grew when Charlotte joined in 2004, bringing the total to 30 teams. Two new franchises, expected to cost **$4–7 billion** each, pushing the count to 32.

Adding two teams forces a full redesign of conferences and divisions. New geometry changes **travel distance, player fatigue, competitive fairness,** and revenue.

Without a model, the decision is driven by politics and lobbying rather than evidence. The paper builds two complementary optimization models to evaluate every option.

15
SCENARIOS
city-pair combinations

02
MODELS
distance · bargaining

32
TEAMS
post-expansion

SIX CANDIDATE CITIES · WEST → EAST

01	VANCOUVER	BC · CANADA	49.3°N
02	SEATTLE	WA · USA	47.6°N
03	LAS VEGAS	NV · USA	36.2°N
04	MEXICO CITY	CDMX · MÉXICO	19.4°N
05	TAMPA	FL · USA	27.9°N
06	MONTREAL	QC · CANADA	45.5°N

MEASURING TRAVEL

THEORY DISTANCE: HOW THE PAPER MEASURES A SEASON OF TRAVEL.

THEORY DISTANCE FOR TEAM *I*

$$\sum_{j \in D_i} d_{ij} \cdot n_1 \quad + \quad \sum_{j \in C_i \setminus D_i} d_{ij} \cdot n_2 \quad + \quad \sum_{j \notin C_i} d_{ij} \cdot n_3$$

n_1 division games

n_2 conference games

n_3 cross-conference

Real schedules churn season to season. The paper estimates travel from league structure alone: division, conference, and game counts, producing a stable, comparable proxy.

TABLE 1 · ANOVA VALIDATION AGAINST ACTUAL NBA TRAVEL, 2004–2018

VARIABLE	F-VALUE	P-VALUE	SIG.
Theory distance	163.3	< 0.001	★★★
# away trips	9.1	0.003	★★
Max trip length	23.6	< 0.001	★★★

AVERAGE DISTANCE TO ALL 30 NBA TEAMS

01	TAMPA	1,168 mi
02	MONTREAL	1,254 mi
03	LAS VEGAS	1,366 mi
04	MEXICO CITY	1,590 mi
05	SEATTLE	1,716 mi
06	VANCOUVER	1,783 mi

MIN-DISTANCE MODEL

MODEL 1: MINIMIZE TOTAL TRAVEL ACROSS ALL 32 TEAMS.

Imagine every team flying to the two new cities each season. Model 1 finds the pair that makes the **total flying distance** across all 32 teams as small as possible.

OBJECTIVE

minimize

$$\Sigma I \in T \; RI$$

total theory travel summed across all 32 teams

$xij \in \{0,1\}$	teams i, j share a division
$yij \in \{0,1\}$	teams i, j share a conference
$ri \geq 0$	theory travel distance for team i

KEY CONSTRAINTS

C(2) **Travel definition.** $ri = \Sigma_j \, dij(n_1 xij + n_2(yij-xij) + n_3(1-yij))$

C(3,4) **One-and-only-one.** Every team belongs to exactly one division and one conference.

C(5,6) **Transitivity.** Same-division and same-conference relationships must be internally consistent, with no broken triangles.

C(8) **Market exposure floor.** $\Sigma_j \, mj(\ldots) \geq \gamma \cdot \bar{M}i$ with $\gamma = 0.8$; every team retains at least 80% of league-average market reach.

NASH BARGAINING MODEL

MODEL 2: NASH BARGAINING FIXES THAT.

Model 1 minimizes total miles, but one team can still get crushed. Model 2 borrows from **game theory** to balance travel efficiency and market revenue so no single team gets stuck with a disproportionate travel burden.

OBJECTIVE
minimize

$$\Sigma i \log(wi)$$

equivalent to maximizing the product of team utilities: the Nash bargaining solution.

wi ≥ 1	travel distance increase ratio (TDIR)
wi = actual ÷ ideal	how much more this team flies vs. its best-case

M1 VS M2: WHAT EACH ONE OPTIMIZES

	MODEL 1	MODEL 2
OBJECTIVE	min total miles	min log-product of TDIRs
FAIRNESS	not considered	✓ **built into objective**
MARKET REVENUE	constraint only	in objective
EFFICIENCY LOSS	none	tiny (~0.001%)
BEST PAIR	TAMPA + MONTREAL	MONTREAL + MEXICO CITY
2ND BEST	Las Vegas + Tampa	Seattle + Montreal

Model 2 picks **Mexico City** : large market, farther away, where Model 1 picks **Tampa**: closer, smaller market. The two objectives work against each other.

ALL FIFTEEN SCENARIOS

TAMPA SHOWS UP IN 4 OF THE TOP 6 PAIRS.

#	CITY 1	CITY 2	MILES	RELATIVE
01	TAMPA	MONTREAL	72,659	
02	LAS VEGAS	TAMPA	76,015	
03	LAS VEGAS	MONTREAL	78,596	
04	TAMPA	MEXICO CITY	82,754	
05	MONTREAL	MEXICO CITY	85,335	
06	SEATTLE	TAMPA	86,524	
07	VANCOUVER	TAMPA	88,517	
08	LAS VEGAS	MEXICO CITY	88,691	
09	SEATTLE	MONTREAL	89,105	
10	MONTREAL	VANCOUVER	91,098	
11	SEATTLE	LAS VEGAS	92,461	
12	LAS VEGAS	VANCOUVER	94,454	
13	SEATTLE	MEXICO CITY	99,200	
14	VANCOUVER	MEXICO CITY	101,193	
15	SEATTLE	VANCOUVER	104,963	

HEADLINE

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Tampa appears in **four of the six** most-efficient pairings. Tampa is the strongest geographic fit of any candidate city.

GEOGRAPHIC LOGIC

Pairings that bracket the league **east-and-west** outperform same-side combinations like Seattle + Vancouver, where everyone has to fly the long way.

SPREAD

Best vs worst differs by **32,304 miles** per season. That gap is a real operating cost, locked in for the life of the franchise.

TOP-FIVE NASH PAIRINGS

ADD MARKET SIZE TO THE MODEL & THE WINNER SHIFTS.

#	CITY PAIR	NASH SCORE	TOTAL MI.	VS M1
01	MONTREAL + MEXICO CITY	0.5525	85,335	↑ from #5
02	SEATTLE + MONTREAL	0.5389	89,105	↑ from #9
03	SEATTLE + MEXICO CITY	0.4750	99,200	↑ from #13
04	MONTREAL + TAMPA	0.4354	72,659	↓ from #1
05	TAMPA + MEXICO CITY	0.3838	82,754	↓ from #4

Higher Nash score → better balance of efficiency and market reach. Distant cities with large markets climb; close cities with smaller markets fall.

THE TWO BIG MOVERS

↑ MEXICO CITY

Rises from #5 in M1 to #1 in M2. Massive metropolitan market (score 0.95, the highest of any candidate) outweighs its distance penalty.

↓ TAMPA

Falls from #1 in M1 to #4 in M2. Geographically ideal but a moderate market (0.55) means other pairs become more attractive when revenue counts.

KEY FINDING

THE TWO OBJECTIVES DIRECTLY CONFLICT.

Both models agree on Montreal but disagree on the second pick. That is the decision the NBA must make.

VERDICT

BOTH MODELS AGREE ON MONTREAL.

MODEL 1 MIN TRAVEL

TAMPA & MONTREAL

Tampa appears in four of the top six scenarios and has the lowest average distance to all 30 existing arenas. Montreal fits the eastern corridor alongside Boston, New York, and Toronto.

TOTAL: **72,659 MI** · -3.4K VS NEXT-BEST · EFFICIENCY LOSS **none**

MODEL 2 NASH BARGAINING

MONTREAL & MEXICO CITY

When market size matters, Mexico City's metropolitan population outweighs the extra miles. Travel cost goes up by only ~0.001% — essentially nothing — while revenue potential improves meaningfully across all 32 teams.

NASH: **0.5525** · EFFICIENCY LOSS **≈ 0.001%**

UNIVERSAL PICK

MONTREAL FEATURES IN BOTH.

The only city that performs well under both models: close enough for distance, large enough for markets.

NBA

THANK YOU.

Two models. Six cities. One decision the league hasn't faced in a generation.

M1 · MIN DISTANCE

TAMPA + MONTREAL

72,659 TOTAL MILES

M2 · NASH BARGAINING

MONTREAL + MEXICO CITY

NASH SCORE 0.5525

SPEAKERS	Prince Chinyama and Kamalesh Kumar
COURSE	DS 808 · Optimization Methods II
PAPER	Fairness, Travel & Market Potential: An Optimization Framework for NBA Expansion
AUTHORS	Hassanzadeh, Davari & Goossens (2025)
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TENSION	travel efficiency ↔ market revenue
AGREEMENT	Montreal, in both models